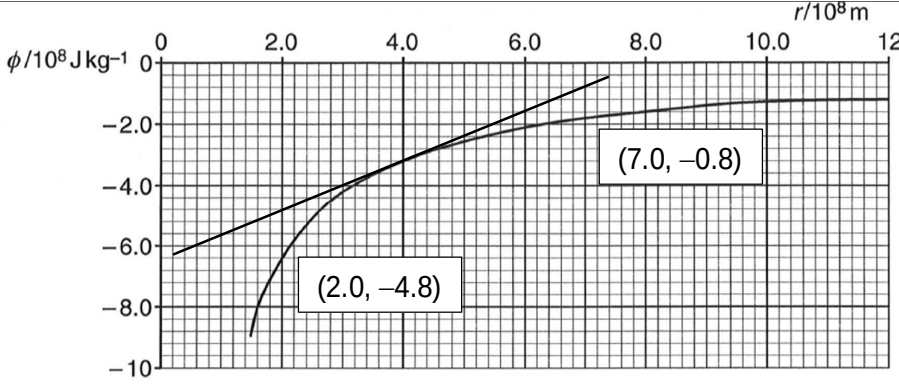


Qns	Answer	Marks
5(a)	The gravitational field strength at a point is the gravitational force (of attraction) per unit mass (acting on a small test mass) placed at that point in the field.	B1
(b)	Gravitational force provides the centripetal force. $G \frac{Mm}{r^2} = m \frac{v^2}{r}$ $G \frac{Mm}{r} = mv^2$ $\frac{1}{2}mv^2 = \frac{1}{2} \times G \frac{Mm}{r}$ Since $E_p = -G \frac{Mm}{r}$, $E_k = -\frac{E_p}{2}$	M1 M1 clear algebra showing proof A0
(c)(i)	From graph, $\phi = -3.2 \times 10^8 \text{ J kg}^{-1}$ at $r = 4.0 \times 10^8 \text{ m}$. $E_{\text{tot}} = E_k + E_p = -\frac{E_p}{2} + E_p = \frac{E_p}{2} = \frac{m\phi}{2}$ $= \frac{8.93 \times 10^{22} \times (-3.2 \times 10^8)}{2}$ $= -1.43 \times 10^{31} \text{ J}$	M1 M1 A1

(c)(ii)	1. From (b), $E_k = G \frac{Mm}{2r}$. Hence, when r decrease, kinetic energy increases 2. From (c)(i), $E_{\text{tot}} = \frac{E_p}{2} = -G \frac{Mm}{2r}$. Hence, when r decrease, total energy decreases.	B1 Show eqn or state relations B1
(c)(iii)	 Method 1 (preferred as it is valid in all cases) Using (2.0, -4.8) and (7.0, -0.8), $g = -\frac{d\phi}{dr}$ $= \frac{[-4.8 - (-0.8)] \times 10^8}{(2.0 - 7.0) \times 10^8} \text{ (minimum 2 s.f.)}$ $= 0.80 \text{ m s}^{-2}$ Method 2 (discouraged; valid only in the case when Jupiter is the only planet contributing to the potential being graphed i.e. the $\phi - r$ graph is an inverse r graph, else the relationship between potential and distance does not hold) $g = \frac{\phi}{r}$ $= \frac{-3.2 \times 10^8}{4.0 \times 10^8}$ $= -0.80 \text{ m s}^{-2}$	M1 A1

Qns	Answer	Marks
6a	The <i>increase</i> in internal energy of a system is equal to the sum of the <i>heat</i>	B2